VNEMOS: Vietnamese Speech Emotion Inference Using Deep Neural Networks

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***Abstract*—This research introduces the dataset that we created to test voice emotional recognition models with Vietnamese data. The data set is the result of research, testing, and filtering 250 emotional segments from movie, movie series and live show divided equally for 5 basic emotional states of humans: "anger, happiness, sadness, neutral and anxiety", VNEMOS contains approximately 30 minutes long. This dataset brings a balanced and diverse emotional set to study emotional recognition issues related to understanding human mood and emotions. Through evaluation Our dataset achieved a 89% accuracy, showcasing the adeptness at capturing the essence of positive emotions and robust learning capabilities in real-world scenarios. By clarifying these nuances. Our research will contribute much to future work on emotion recognition and signal processing in Vietnamese and data using human psychological science. Detailed information of the dataset is in the following link: "bit.ly/VNEMOS"**

***Index Terms*—Vietnamese, Speech emotions dataset, Speech emotion recognition, Cognitive science, Signal processing**

I. Introduction

Over the past decades, many studies have been conducted to study the human ability to recognize and analyze emotions. Therefore, emotion datasets play an important role in developing emotion recognition models. Emotions play an important role in developing models related to human emotion recognition, or developing AI to have emotions or actions like humans. Through this, it will help humanity increasingly develop towards a developed society with AI machines that can help support more people and reduce the amount of work a person has to endure as well as help reduce failure. Emotion datasets are being devised for various research and applications, providing an avenue to comprehend and quantify human emotions. Throughout human cognitive development, behavior and social interactions, emotions play a crucial role in determining decision-making processes, communication and patterns. Research has demonstrated that cultural nuances profoundly shape emotional expression and perception. This suggests the need to create an emotion dataset suitable for a specific cultural context like Vietnam [1]. A suitable dataset will help accurately capture and analyze the emotions of each individual in the Vietnamese. These datasets, moreover, represent imperative resources, serving as the foundation for the development of applications, ranging from emotional computing systems to virtual agents [2]. By addressing the unique emotional in the Vietnamese, these initiatives not only advance scientific understanding but also facilitate the development of more inclusive and effective technologies. Moreover, these developments cater to a diverse range of user groups. Emotion recognition in human-computer interaction is very important, however most existing datasets are mainly in English, creating an emotion recognition dataset in Vietnamese will help researchers be able to study emotion recognition in this language [3]. In this research, we build a Vietnamese speech emotion dataset through movies and live shows. The Vietnamese Emotion Dataset is developed to solve the problem of lack of data for research on emotion recognition in Vietnamese.

In recent years similar studies have been conducted on the development of speech emotion datasets. First, we must mention IEMOCAP [4] and MELD [5] as two of the many large databases that we have access to. DAIC-WOZ is an equally famous dataset, which is a relatively new interview voice database [6], developed by the University of Southern California. There are currently about 200 interview recordings, which range in length from 7 minutes to 33 minutes (on average, 16 minutes). There are several types of accompanying data, including verbatim transcripts, timestamps at the beginning and end of sentences, facial expressions, and eye contact information, in addition to audio recordings. The IEMOCAP dataset [4] consists of 7,433 sentences, totaling 13 hours and 40 minutes, spoken by 10 individuals, while the MELD dataset [5] consists of 13,708 sentences, about 12 hours, from 407 individuals. The application of machine learning methods has enhanced the quality of several datasets. Schuller et al. utilized a continuous Hidden Markov Model (HMM) for Speech Emotion Recognition (SER), constructing a self-collected speech corpus [7]. At the IEMOCAP and AIBO datasets, decision trees have been deployed to minimize error propagation. Alternatively, as seen with the EmoDB and ETERFACE datasets, RBM has been applied to learn discriminative features. With the rapid development of deep learning, Zhang et al. used DCNN to bridge the emotional gap in speech signals on the RML dataset [8], [9]. Also recently, Yeh et al. [10] proposed a dialogue emotion decoding algorithm to continuously decode the emotional states of each utterance on IEMOCAP [4] and MELD [5]. Detailed information of related studies is shown in Table I.

II. Vietnamese Speech Emotion Dataset

In this study, we develop a Vietnamese speech emotion dataset, the proposed dataset is a mixed dataset between acted and natural. The samples were collected from movies, series

TABLE I: Research on Emotions Speech Datasets

Dataset	Language	Type of Dataset	Information	Emotions
FAU Aibo	German	Natural	9 hours of German speech of 51 children at the age 10 -13 years interacting with pet	11 (not specified)
EMODB	German	Acted	The data set contains 533 tracksets from five actors, five actresses	Anger, neutral, fear, boredom, happiness, sadness, disgust
KES	Korean	Acted	5400 utterances from 10 actors	Neutral, joy, sadness, anger
RML	English, Mandarin, Urdu, Punjabi, Persian, and Italian	Acted	Collected video samples from eight subjects, 500 video samples. Recorded at a sampling rate of 22050 Hz, using channel 16-bit	Happiness, sadness, anger, fear, surprise, disgust
IEMOCAP	English	Acted	Approximately 12 hours of audiovisual data included in the document. Recorded data from ten participants	Anger, excitement, frustration, happiness, sadness, neutral
Drivers Stress	English	Natural	598 utterances with the average length of an utterance being 1.6 seconds	Neutral, Stress
MELD	English	Acted	13,000 utterances from 1,433 dialogues from the TV-series Friends	Anger, disgust, fear, joy, neutral, sadness, surprise
VNEMOS (Our)	Vietnamese	Acted, Natural	250 segments, approximately 30 minutes from 27 movies, movie series and live shows	Anger, happiness, sadness, neutral, anxiety

and live shows, most part of the emotional sentences were collected from professional actors expressing them, a small part of were actual emotions from the live shows. We name this dataset Vietnamese Emotion Speech Dataset (VNEMOS).

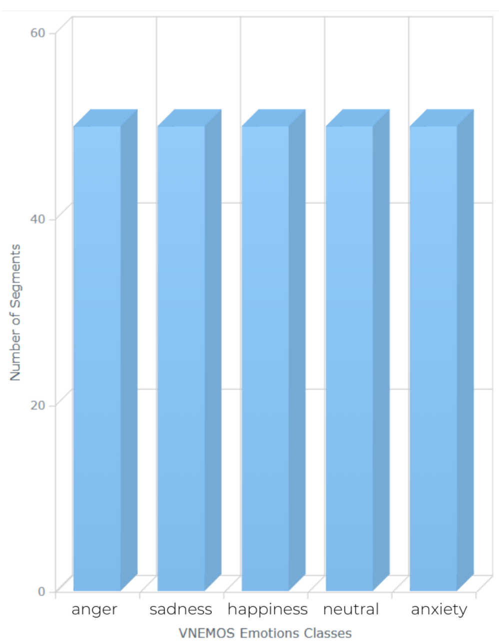

Fig. 1. Emotions Classes Distributions in VNEMOS.

In VNEMOS, segment emotional sentences are saved as wav files. The dataset is composed of five basic human emotions: happiness, sadness, anger, neutral and anxiety. Factors such as the natural environment and background noise of vehicles, animals and background music of programs have been filtered and removed to increase the accuracy of the proposed dataset. And the clips must ensure all elements such as: no background music, only contain one type of emotion, must be clearly emotional. being filtered once, the videos filtered again and converted into WAV audio format. WAV files will also be filtered to avoid possible corrupted or lost WAVs. In the dataset, each emotion is collected in 50 segments, the dataset visualization diagram is shown in Figure 1. There are 27 types of programs including “1 live show, 11 movies, 15 movie series” Figure 2 illustrates the data source distribution from entertainment programs have been collected for the dataset. All films have been carefully evaluated and selected with actors with good acting and confident performance styles that are well known in the industry. In the collecting process, all of the data must meet the following conditions:

- Sound must be clear.
- Characters must express emotions that contain only one emotion.
- The emotion segments must be identifiable.

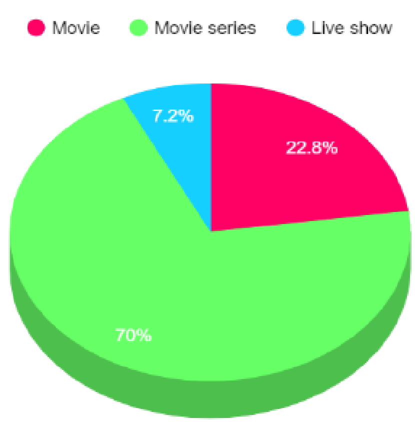

Fig. 2. Data Source Distribution from Entertainment Programs.

Figure 3 illustrates the age distributions the data we have collected, the emotions will be mainly collected from the age of 20 - 50 or the age over 50. Because perhaps under the age

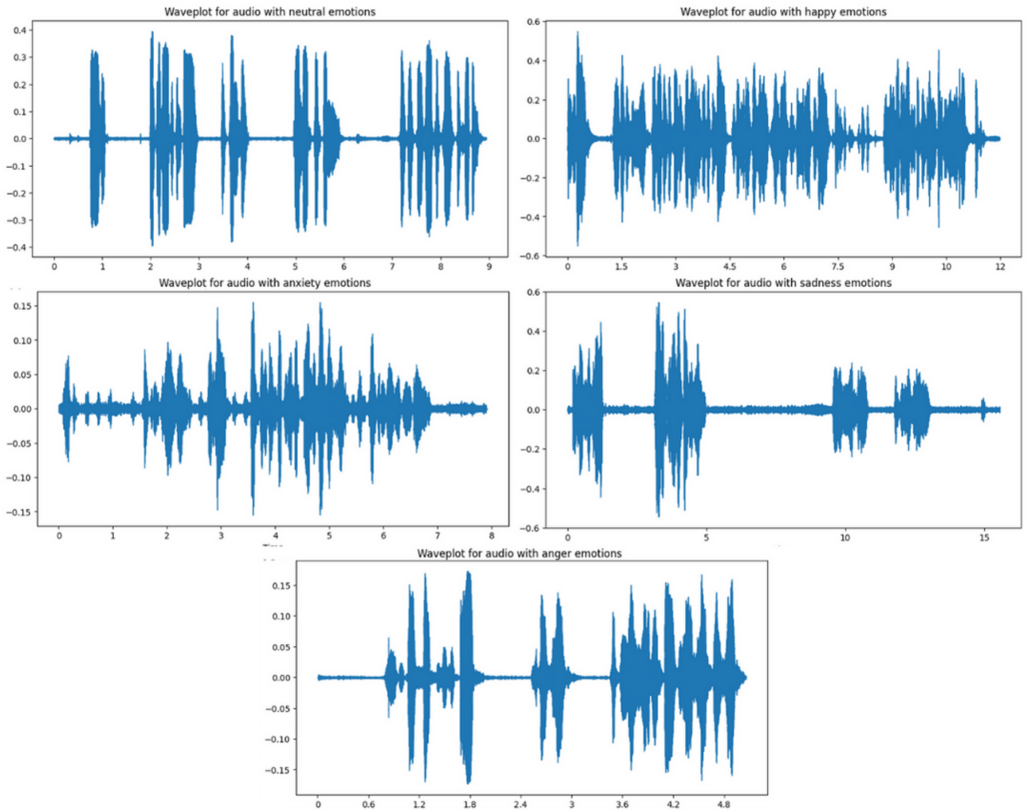

Fig. 4. Speech Emotions Sound Wave from Five Classes in VNEMOS with x Axis for Times and y Axis for Amplitude.

of 20 is the age that we are still. If your emotions are not yet stable, even though there are many emotions expressed at that age, those emotions may not be clear or those emotions maybe mixed with other emotions, causing confusion difficulty in collecting and filtering data. As for other ages, they have certain maturity in their emotions and life experiences, so it is understandable that the number in the remaining two age groups is higher, and where the age is from 20 - 50 is the most because at that time people have to face life, hardships and even the smallest joys in life will be recreated on screen in the most realistic way.

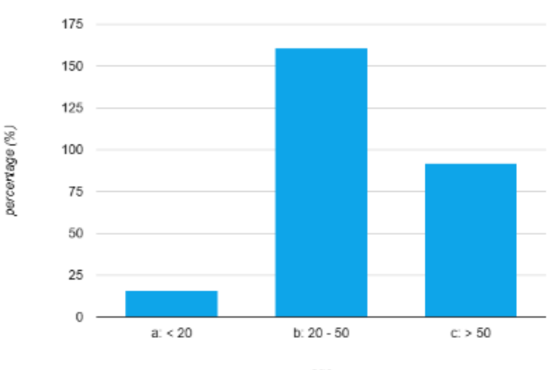

Fig. 3. VNEMOS Age Distributions.

Also, Figure 4 illustrates the emotions sound wave that we have converted after collecting data and filtering to be able to produce the most realistic emotions data. Figure 5 illustrates VNEMOS Emotions data length distribution, for each chart is a different emotion source from male, female and lastly in total length of the dataset.

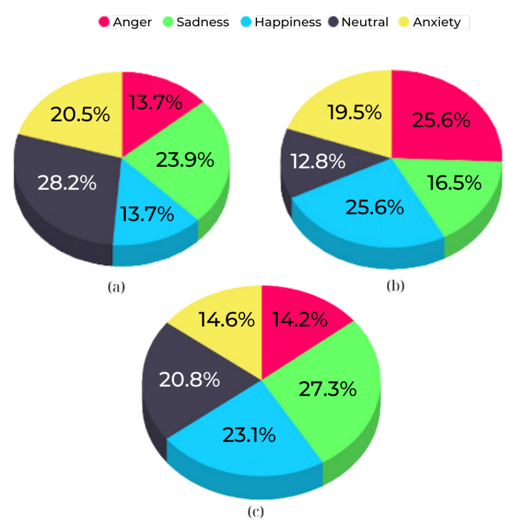

Fig. 5. VNEMOS Emotions Data length Distribution (a) Male, (b) Female, (c) VNEMOS.

The total number of seconds for all emotions is approximately 30 minutes long, more specifically 1562.29 seconds with anger is 14.2%, sadness is 27.3%, happiness is 23.1%, neutral is 20.8% and anxiety is 14.6%. Because the emotions of anxiety in movies or on screen will mainly consist of

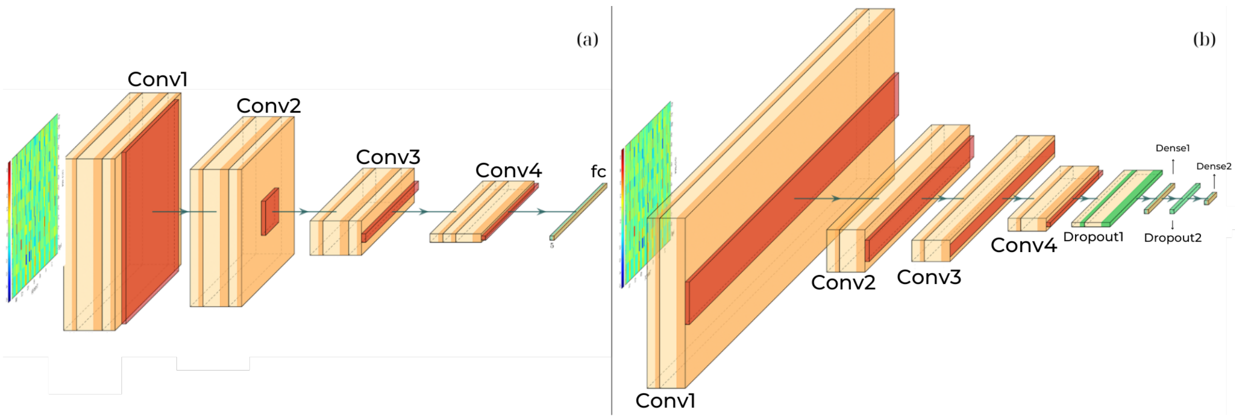

Fig. 6. CNN Models Structure (a) Our Model, (b) Gideon Model.

screams and screams, and if there is dialogue, it will likely include another type of emotion, which could be anger, sad. Therefore, the emotion of anxiety is often more difficult to capture than other emotions. On the contrary, compared to anxiety, sadness can be said to be the easiest to capture because at this time the character is in a mood that can almost be said to be hopelessness, suffering, fatigue, regardless of everything, so this emotion is more than the rest, which is understandable and this increase is not significant, so it will not affect the diversity of the dataset.

III. Evaluation experiments

To evaluate the proposed dataset, we design a SER model using tone and pitch through a 2D Convolutional Neural Networks (CNN) network using input data processed as Mel Frequency Cepstrum (MFCC). Input data is also preprocessed by reducing them to 5 seconds and adjusting them to 16kHz. Data segments that do not have sufficient input time will also be filled in using mirror padding technique. Upon entering training, the data is cross-validated in order to divide the amount of data used for each work and the outcome of this model is the Fully Connected (FC) class composed of five classes: anger, sadness, happiness, neutral, and anxiety. Additionally, we also use Gideon Oboro's SER Model [15] to create diversity in evaluating results, the Model use 1D CNN network, the input data get preprocessed by adding noise, stretching and pitch, increasing and decreasing data speed, and also processing the input data as MFCC. Figure 6 provides a schematic representation of the models, Table II and III shown the models structure that we use in this research with 294085 $params$ from our model and 412421 $params$ in Gideon model.

In this research, we trained two models, each model was trained over 100 epochs on Google Colab using High RAM Colab Runtime and NVIDIA T4 GPU. Table IV and V outlined confusion matrix after training and validating.

To evaluate the dataset, we used four metrics, including Precision (P), Recall (R) and F1 - score as described in Equation 1.

TABLE II: *Our CNN Model Structure*

Layers	OutputDim	Kernal	Stride
2D conv	128×157×64	3×3	1×1
2D pooling	64×78×64	2×2	2×2
2D conv	64×78×64	3×3	1×1
2D pooling	16×19×64	4×4	4×4
2D conv	16×19×128	3×3	1×1
2D pooling	4×4×128	4×4	4×4
2D conv	4×4×128	3×3	1×1
2D pooling	1×1×128	4×4	4×4
flatten	128	-	-
fc	5	-	-

TABLE III: *Gideon CNN Model Structure*

Layers	OutputDim	Kernal	Stride
1D conv	40×256	7	1
1D pooling	20×256	2	2
1D conv	20×128	7	1
1D pooling	10×128	2	2
1D conv	10×128	7	1
1D pooling	5×128	2	2
1D conv	5×64	7	1
1D pooling	3×64	2	2
flatten	32	-	-
fc	5	-	-

TABLE IV: Confusion Matrix of Classifying Emotions in VNEMOS from Our Model

	Anger	Anxiety	Happiness	Sadness	Neutral
Anger	15	0	0	0	0
Anxiety	0	14	1	0	0
Happiness	0	0	15	0	0
Sadness	0	1	6	8	0
Neutral	0	0	0	0	15

TABLE V: Confusion Matrix of Classifying Emotions in VNEMOS from Gideon Model

	Anger	Anxiety	Happiness	Sadness	Neutral
Anger	5	1	1	1	0
Anxiety	1	4	0	2	2
Happiness	2	1	11	0	1
Sadness	0	0	2	8	0
Neutral	0	0	0	0	8

$$\begin{cases} \text{Recall} = \dfrac{TP}{TP+FN} \\ \text{Precision} = \dfrac{TP}{TP+FP} \\ \text{F1-score} = 2 \times \dfrac{P \times R}{P+R} \end{cases} \quad (1)$$

where TP indicates the number of correct prediction, FP represents the number of incorrect prediction, and FN gives the number of cases has not detected by that the model.

The VNEMOS dataset contains approximately 30 minutes of data divided into 250 segments based on five different emotions, despite the relatively short duration of the dataset, it demonstrated impressive performance after training and validating from both models, demonstrating the high quality and richness of the data. Table VI and VII outlined the evaluations from two models.

TABLE VI: VNEMOS Evaluations Results from Our Model

	Precision	Recall	F1-score
Anger	1.00	1.00	1.00
Anxiety	0.93	0.93	0.93
Happiness	0.68	1.00	0.81
Sadness	1.00	0.53	0.70
Neutral	1.00	1.00	1.00
Accuracy			0.89

TABLE VII: VNEMOS Eevaluations Results from Gideon Model

	Precision	Recall	F1-score
Anger	0.62	0.62	0.62
Sadness	0.73	0.80	0.76
Happiness	0.79	0.73	0.76
Neutral	0.73	1.0	0.84
Anxiety	0.67	0.44	0.53
Accuracy			0.72

As a result of using 2D data as input features, our CNN model achieved validation accuracy at 89%, which indicates the potential for the dataset for development, where it is very remarkable how well it can distinguish complex patterns in a limited dataset. In Gideon model has achieved 72% accuracy, the model robustness against over-fitting along with its competitive accuracy metrics, highlight the dataset's potential in facilitating the training of models capable of effectively recognizing emotions from voice. While the brevity of the dataset poses inherent challenges, its ability to deliver compelling results emphasizes its quality and effectiveness.

IV. Conclusion

Our research introduced the VNEMOS dataset, an emotional data set in Vietnamese. The dataset is collected voices from three typical regions of Vietnam: the North, Central, and South, with different age groups, for the purpose of identifying and understanding the psychology and emotions of people. Our research was done with a small set of data with 50 segments per emotion, achieving accuracy of 89% from validation. Although our research hasn't actually achieved an optimal outcome, with such a small set of data, this is a positive indicator, and this has laid the foundation for Our future research. In the future, we have detailed plans to develop this project and, above all, increase the emotional quantity of the data set to produce better results. Besides, we expect to use more facial features than speech to make input data so that we can understand human psychology, cognitive behavior and emotions more accurately and clearly.

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